CSE 111 Fall 2026: Project Checkpoint 1

YouTutor: Campus AI Learning Platform

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Synopsis

YouTutor is an AI-powered educational application designed to help university students learn course material through personalized tutoring and interactive practice. Students can select courses and topics, ask an AI tutor questions, receive explanations, and complete AI-generated practice problems. Unlike traditional tutoring platforms, the system does not rely on human tutors and instead provides an interactive, self-paced learning experience.

Behind the scenes, a relational database stores students, courses, topics, learning sessions, AI conversations, practice questions, and performance data. Because it keeps a record of each student's history, question attempts, and accuracy, the platform can spot the topics a student is struggling with and serve up more practice on those. The idea is that the experience adjusts to each student as they progress.

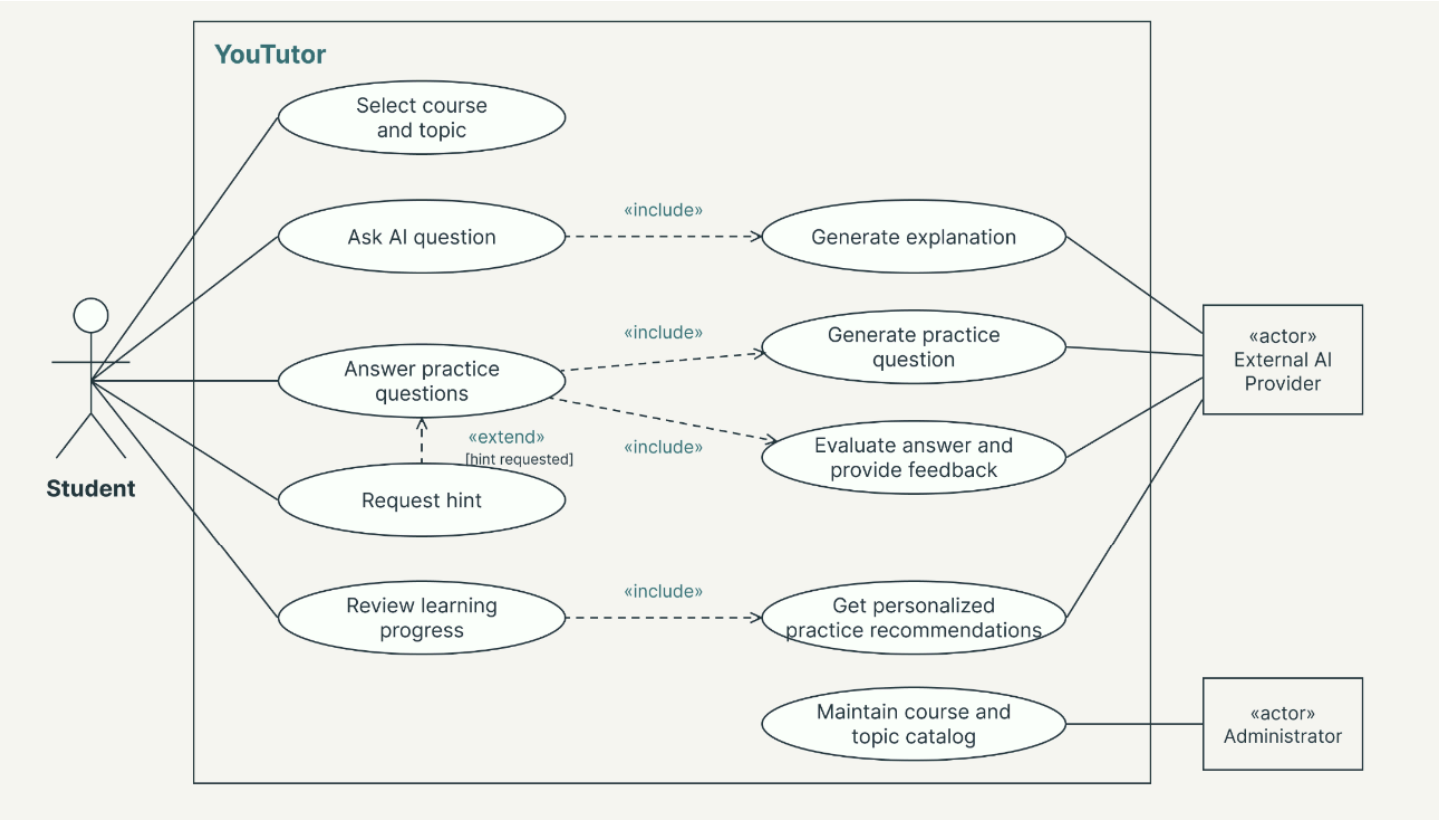
Authors & Contribution: Julian Lee: 50%, Richard Chen: 50%

UML Use Case Diagram

[Insert description of your main use case(s), i.e., write plain text describing the main use case.]

Authors & Contribution: [Provide the contribution % of each team member]

Use Case Diagram:

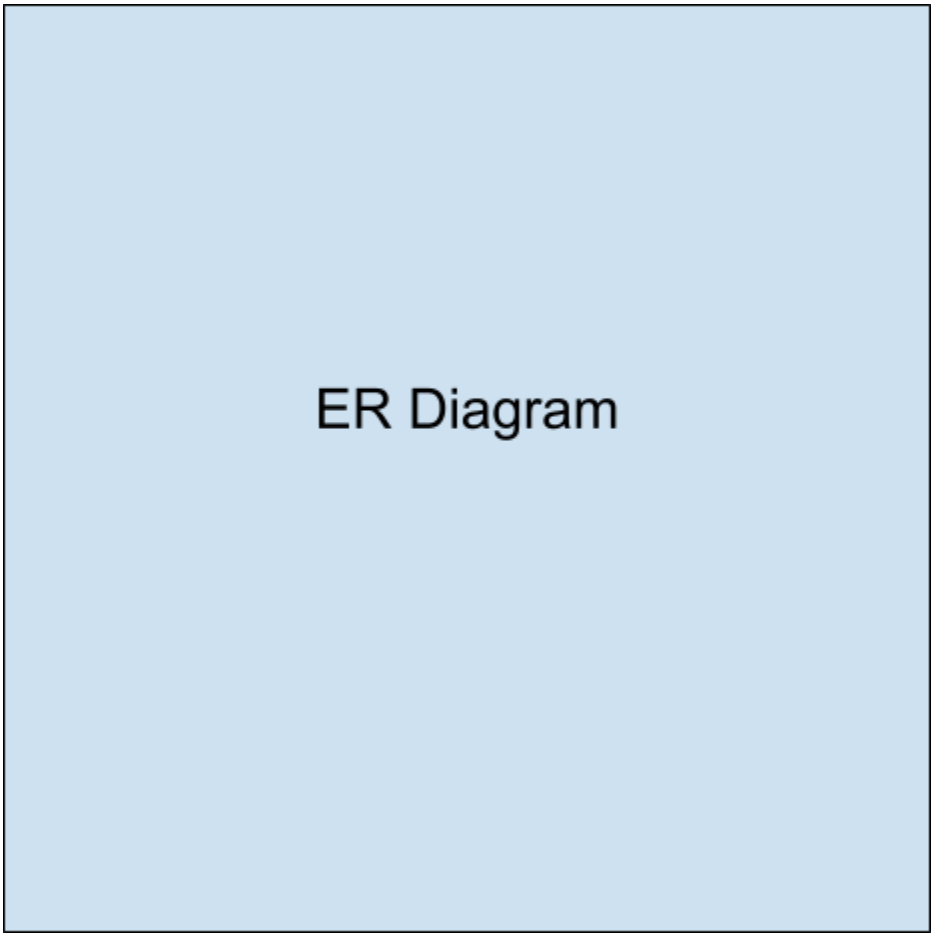


Authors & Contribution: Julian Lee 50% and Richard Chen 50%

ER Diagram

[Describe your ER diagram(s) here. The minimum requirement is 6 entities and 6 relationships (out of which at least 2 many-many).]

ERD-1:



Authors & Contribution: [Provide the contribution % of each team member]

(Add more ER diagrams as needed and number them sequentially with contribution listed under each)

Relational Schema

[Describe your database's relational schema here]

[Relation 1]

[Insert schema]

Authors & Contribution: [Provide the contribution % of each team member]**:**

[Relation 2]

[Insert schema]

Authors & Contribution: [Provide the contribution % of each team member]**:**

(Add more relations and their schema as needed and number them sequentially with contribution listed under each)

NOTE: Name your file **Checkpoint1-specs** and submit it as a PDF.